House Price Prediction: Project Review Report

Source: project notebook (Google Colab) | Stage reviewed: EDA and preprocessing | Prepared for review

1. Summary

This project prepares a house price prediction dataset for modelling. The notebook loads the data, runs exploratory data analysis (EDA), generates an automated profiling report, builds a preprocessing pipeline, creates a train/test split and starts outlier checks. **No model has been trained or evaluated yet.**

The main finding is that the dataset appears to be synthetic and that Price shows almost no visible relationship with any feature. That limits how well any model can perform. The notebook also has several small code issues, listed in section 5, which should be fixed before the next stage.

2. Dataset overview

Item	Detail
Size	2,000 rows, 10 columns
Target	Price (integer)
Numeric features	Area, Bedrooms, Bathrooms, Floors, YearBuilt (plus Id, an identifier)
Categorical features	Location (Downtown, Urban, Suburban, Rural), Condition, Garage (Yes/No)
Missing values	None in any column
Duplicate rows	0

Location is balanced across the four categories: Downtown 558, Urban 485, Suburban 483, Rural 474.

Key statistics

Variable	Min	Mean	Max
Price	50,005	537,677	999,656

Variable	Min	Mean	Max
Area	501	2,786	4,999
YearBuilt	1900	1961	2023

Bedrooms range from 1 to 5, Bathrooms from 1 to 4 and Floors from 1 to 3. Price has a standard deviation of about 276,000.

3. What the notebook does

1. **Inspection:** `info`, `shape`, `head`, `sample`, `describe`, missing-value and duplicate checks.
2. **Visual EDA:** Location count plot, Price histogram and KDE, box plots, an Area vs Price scatter plot and a Location vs Price bar plot.
3. **Automated EDA:** a `ydata-profiling` report exported to HTML.
4. **Preprocessing:** Price is separated as the target and Id is dropped. Numeric columns are scaled with `StandardScaler`, and categorical columns are encoded with `OneHotEncoder(drop="first")` inside a `ColumnTransformer`.
5. **Split:** 80/20 train/test with `random_state=42`, giving 1,600 training and 400 test rows. Each has 12 transformed features (5 numeric plus 7 one-hot columns).
6. **Outlier analysis:** IQR bounds for Price were computed as about -419,884 to 1,500,068 (IQR about 479,988).

The preprocessing design is sound. The encoder and scaler are fit on the training data only, which avoids data leakage.

4. Findings

The data looks synthetic. The Price histogram is roughly flat between about 50,000 and 1,000,000. Its mean (about 538k) and standard deviation (about 276k) match what a uniform distribution over that range would give. Real housing prices are usually right-skewed.

Price shows no visible relationship with features. The Area vs Price scatter plot is an even cloud with no trend. Median price is similar across all four Location categories (roughly 520k to 560k, with overlapping error bars). Larger houses or better locations do not appear to cost more.

There are no outliers in Price. The IQR upper bound (about 1.5M) is above the maximum price (about 1.0M) and the lower bound is negative, so no value falls outside the limits. No trimming is needed.

Consequence for modelling. If the features carry little signal, expect a model to explain very little variance (R^2 near zero). That would be a property of the data, not necessarily a modelling error.

5. Issues to fix

#	Issue	Suggested fix
1	`df.corr` is missing parentheses, so it prints the method instead of a correlation matrix.	Use `df.corr(numeric_only=True)` and show it as a heatmap.
2	`df.drop_duplicates()` is not assigned, so it has no effect.	Assign the result. It is harmless here because there are 0 duplicates.
3	The trimming filter uses `|` (OR), so it is true for every row, and the result is neither saved nor applied. The 250,000 and 400,000 bounds are also arbitrary.	Remove it, or use the IQR bounds with `&` and assign the result.
4	Several plots have wrong labels. The Location box plot is titled "House Price Distribution" with y-axis "Density" and x-axis "House Price". The Area box plot and the loop box plots (titled literally "box_plot of col") have the same problem.	Set titles and axis labels to match the variable plotted.
5	`sns.countplot(df["Location"])` and `sns.boxplot(df["Area"])` pass data positionally.	Use `x=` or `y=` keywords for compatibility with newer seaborn versions.
6	`YearBuilt` is scaled as a plain number, and `Id` is only dropped after the split setup.	Consider a derived House Age feature, and drop Id up front.
7	The data path is hard-coded (`/House Price Prediction Dataset (1).csv`), and `ydata-profiling` is flagged as deprecated.	Use a relative path or a Drive mount, and pin the package version.
8	Section headers include an unused "New section" placeholder, and comments contain typos.	Clean up the notebook for presentation.

6. Recommendations and next steps

1. **Confirm the data source.** If the dataset is synthetic, say so in the report, and consider a real dataset such as Ames Housing or a Kaggle housing set for meaningful results.
2. **Check feature relationships properly.** Compute the correlation matrix and group Price by Condition, Garage, Bedrooms and Bathrooms, which have not been explored yet.
3. **Train baseline models.** Start with Linear Regression, then Random Forest and Gradient Boosting, using a `Pipeline` that combines the preprocessor and the model.
4. **Evaluate with cross-validation.** Report MAE, RMSE and R^2 on the test set, plus 5-fold cross-validation, and compare against a mean-price baseline.
5. **Interpret results.** Add feature importance or permutation importance, and discuss honestly if the models barely beat the baseline.
6. **Document the work.** Add short markdown notes explaining each stage and the conclusion drawn from each plot.

7. Overall assessment

Area	Status
Data loading and inspection	Complete
Data quality	Good: no missing values or duplicates
EDA	Good coverage, but labelling errors need fixing
Preprocessing pipeline	Well designed, no leakage
Outlier handling	Checked, none found; the trimming code needs removal
Modelling and evaluation	**Not started**
Documentation	Minimal

The project is a solid foundation for modelling. Before it is ready for final review, the code issues should be fixed, the data's synthetic nature acknowledged and a baseline model with evaluation metrics added.